Assembly practical or visualization practical?



About this practical

Simulated data

- Fully simulated genomes and reads
- Short runtimes: focus on interpretation

Freedom to explore

- Choose your tools and parameters
- **No need to complete every task**
- Stuck? Try another tool

Supporting material

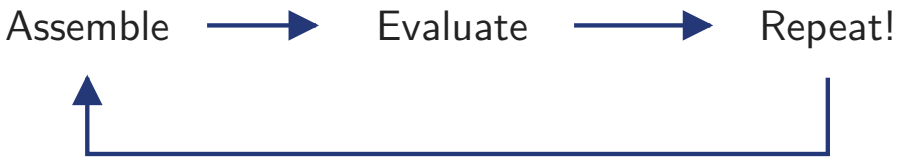
- Cheatsheet: check commands when needed
- `install_party.pdf`: installation without Docker

Investigating the datasets

SeqKit

How many reads and bases are there?
What are the mean read length and N50?
What are the shortest and longest reads?
What is the depth for a genome of approximately 500 kb?

Genome assembly

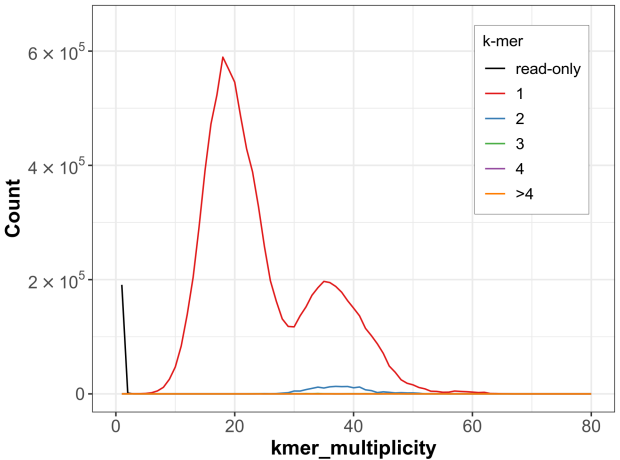


Success?

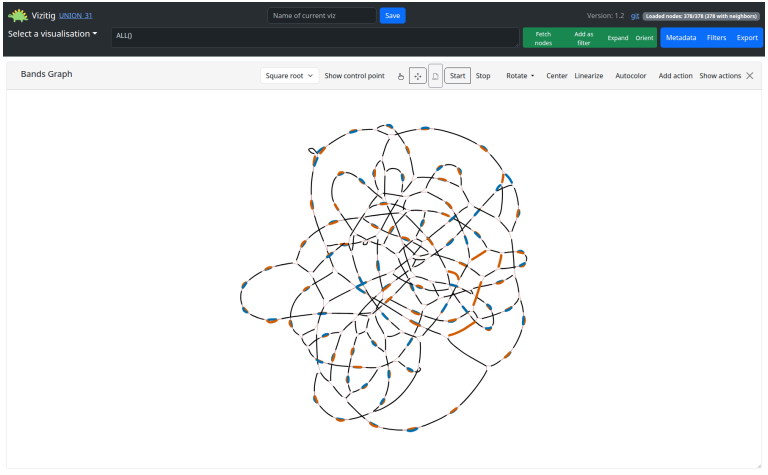
Reference based evaluation: QUAST

Alignment-based statistics	**SPADES**	**HYBRID**	**VERKKO**
Genome fraction (%)	84.504	99.64	89.33
Duplication ratio	1.001	1.02	1.527
Largest alignment	40 031	199 821	104 175
Total aligned length	422 892	507 965	682 022
NGA50	20 062	127 993	97 512
LGA50	9	2	3
Misassemblies			
# misassemblies	4	6	10

kmer based evaluation: MerquryFK



Visualization: Vizitig



SeqKit: read statistics

```
seqkit stats -a R1 R2 HIFI ONT
```

Sample reads	Reads	Bases	Read N50
Illumina, mate 1	20,228	5,057,000	250
Illumina, mate 2	20,228	5,057,000	250
HiFi	631	10,117,819	16,437
Nanopore	188	10,156,363	57,618

Depth = total sequenced bases / genome length.

R1 and R2 together: approximately 20× depth for 500 kb.

SPAdes: pairing information

```
spades.py --only-assembler -k 31 -t 4 -m 8 --s1 R1 --s2 R2 -o
UNPAIRED
```

```
spades.py --only-assembler -k 31 -t 4 -m 8 -1 R1 -2 R2 -o PAIRED
```

k = 31	**Contigs**	**Bases**	**N50**
Unpaired	38	422,893	20,735
Paired end	21	426,453	35,000

`contigs.fasta`: contigs. `scaffolds.fasta`: scaffolds.

Assembly statistics use contigs ≥ 500 bp throughout.

SPAdes: kmer size

```
spades.py --only-assembler --s1 R1 --s2 R2 -k 31 -t 4 -m 8 -o
OUTPUT
```

k	Contigs	Bases	N50
21	38	422,293	23,614
31	38	422,893	20,735
55	39	424,240	20,783
77	39	425,604	20,827
99	39	426,968	20,871
127	43	429,178	18,882

MEGAHIT

```
megahit -1 R1 -2 R2 --k-list 21 --min-contig-len 500 -t 4 -m
8589934592 -o SINGLE_OUT
```

```
megahit -1 R1 -2 R2 --k-list 21,41,61,81,101,121 --min-contig-len
500 -t 4 -m 8589934592 -o MULTI_OUT
```

k	Contigs	Bases	N50
21	99	419,326	9,180
41	85	422,202	10,517
61	77	424,676	12,041
81	69	426,652	12,061
101	61	428,166	19,522
121	54	429,319	14,425
Multi k	51	428,998	19,562

`final.contigs.fa`: assembled contigs.

Flye: HiFi and Nanopore reads

```
flye --pacbio-hifi HIFI --genome-size 500k --threads 4 --out-dir
HIFI_OUT
```

```
flye --nano-hq ONT --genome-size 500k --threads 4 --out-dir ONT_OUT
```

Reads	Contigs	Bases	N50
HiFi	6	365,174	215,768
Nanopore	4	428,443	109,999

`assembly.fasta`: contigs. `assembly_graph.gfa`: repeat graph.

hifiasm: HiFi reads

```
hifiasm -o OUTPUT -t 4 -l0 -f0 HIFI
```

Reads	Contigs	Bases	N50
HiFi	5	492,006	130,321

`OUTPUT.bp.p_ctg.gfa`: primary contigs, with sequences.

`-l0` disables duplicate purging for this haploid sample.

hifiasm: adding ultralong reads

```
hifiasm -o OUTPUT -t 4 -l0 -f0 --ul ONT HIFI
```

Reads	**Contigs**	**Bases**	**N50**
HiFi	5	492,006	130,321
HiFi + Nanopore	1	511,466	511,466

Does one contig establish that the assembly is complete and correct?

Verkko: HiFi and Nanopore reads

```
verkko -d HIFI_OUT --hifi HIFI --no-nano --no-correction
--no-consensus-bam --haploid --local-cpus 4 --local-memory 8
```

```
verkko -d HYBRID_OUT --hifi HIFI --nano ONT --no-correction
--no-consensus-bam --haploid --local-cpus 4 --local-memory 8
```

Reads	**Contigs**	**Bases**	**N50**
HiFi only	9	216,138	37,127
HiFi + Nanopore	9	685,570	97,512
Nanopore only	Unsupported for these reads		

`assembly.fasta`: assembled sequences.

QUAST: comparison with the related reference

```
quast.py SPADES.FA HYBRID.FA VERKKO.FA -r REFERENCE.FA --min-contig
500 --threads 4 -o OUT
```

Alignment-based statistics	SPADES	HYBRID	VERKKO
Genome fraction (%)	84.504	99.64	89.33
Duplication ratio	1.001	1.02	1.527
Largest alignment	40 031	199 821	104 175
Total aligned length	422 892	507 965	682 022
NGA50	20 062	127 993	97 512
LGA50	9	2	3
Misassemblies			
# misassemblies	4	6	10

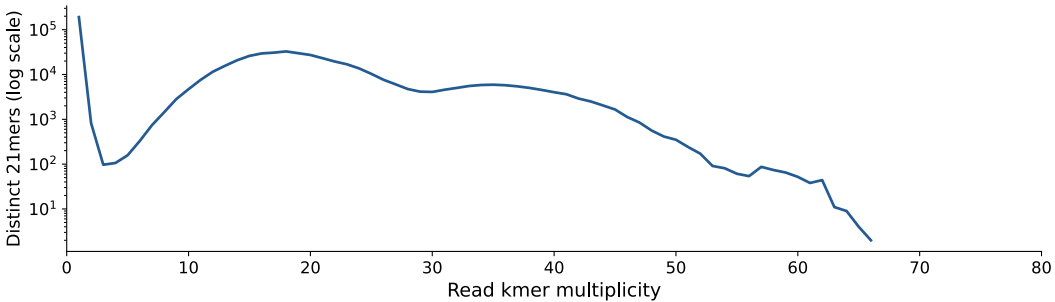
`report.html`: summary. `icarus.html`: alignments.

Reference disagreements can reflect genuine variation.

FastK: count read kmers

```
FastK -k21 -t1 -M2 -T4 -NREADS R1 R2
```

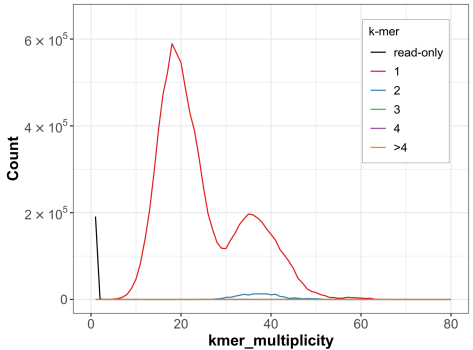
```
Histex -A -h1:500 READS > READS.hist.tsv
```



`READS.ktab`: kmer table. `READS.hist.tsv`: multiplicity histogram.

MerquryFK: copy number spectrum

```
MerquryFK -T4 -X80 READS SPADES.FA EVALUATION
```



SPAdes, k = 31: 100% of reliable Illumina kmers recovered.

MerquryFK: quality and completeness

```
MerquryFK -T4 READS ASSEMBLY.FA EVALUATION
```

Assembly	**QV**	**kmer completeness**
SPAdes, k = 31	∞	100.00%
Flye HiFi	35.6	81.03%
Verkko hybrid	29.2	93.98%

∞: no unsupported assembly kmers at k = 21.

Illumina depth holes can penalise sequence recovered by long reads.

Distinct kmer completeness can miss collapsed repeat copies.

Bandage: the SPAdes assembly graph

```
QT_QPA_PLATFORM=offscreen BandageNG image GRAPH.gfa GRAPH.png
--width 1600
```



SPAdes, k = 31. Native graph: 130 nodes, 172 edges.

GGCAT: kmer size

```
ggcat build -k 31 -s 3 -e -j 4 -m 1 R1 R2 -o GRAPH.FA
```

```
ggcat build -k 63 -s 3 -e -j 4 -m 1 R1 R2 -o GRAPH.FA
```

```
ggcat build -k 127 -s 3 -e -j 4 -m 1 R1 R2 -o GRAPH.FA
```

k	Sample unitigs
31	131
63	107
127	218

GGCAT does not remove tips or collapse bubbles.

GGCAT: assembly and reference union

```
ggcat build -k 31 -s 1 -e -j 4 -m 1 ASSEMBLY.FA REFERENCE.FA -o
UNION.FA
```

Unitigs	Shared	Sample only	Reference only
378	206	84	88

Sample: hifiasm HiFi assembly. Reference: supplied related genome.

`UNION.FA`: unitig sequences with graph links.

Vizitig: create the union from FASTA

```
export VIZITIG_DIR="$PWD"
```

```
mkdir -p "$INPUT_FOLDER/sample" "$INPUT_FOLDER/reference"
```

```
cp ASSEMBLY.FA "$INPUT_FOLDER/sample/ASSEMBLY.fasta"
```

```
cp REFERENCE.FA "$INPUT_FOLDER/reference/reference.fasta"
```

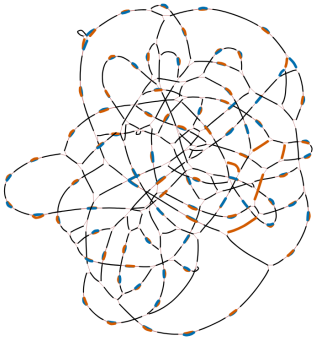
```
vizitig create "$INPUT_FOLDER" -k 31 -f 1 -n UNION
--edge-annotation
```

```
vizitig run --hostname 0.0.0.0 --no-launch-browser
```

Set `INPUT_FOLDER` to a new directory name. Open http://localhost:4242.

Vizitig: complete union graph, k = 31

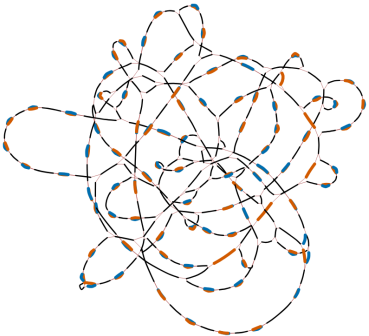
```
vizitig create "$INPUT_FOLDER" -k 31 -f 1 -n UNION_31
--edge-annotation
```



Sample only: blue. Reference only: orange. Shared: black.

Vizitig: complete union graph, k = 63

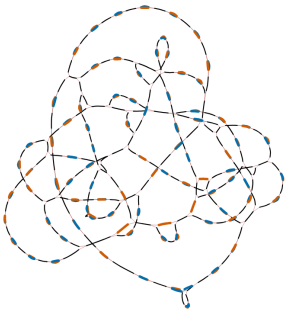
```
vizitig create "$INPUT_FOLDER" -k 63 -f 1 -n UNION_63
--edge-annotation
```



Sample only: blue. Reference only: orange. Shared: black.

Vizitig: complete union graph, k = 127

```
ggcat build -k 127 -s 1 -e -j 4 -m 1 ASSEMBLY.FA REFERENCE.FA -o
UNION.FA
```



Sample only: blue. Reference only: orange. Shared: black. Import and colour `UNION.FA` with Vizitig. See cheatsheet.

Vizitig: a single nucleotide difference

```
vizitig run --hostname 0.0.0.0 --no-launch-browser
```

Score: 500

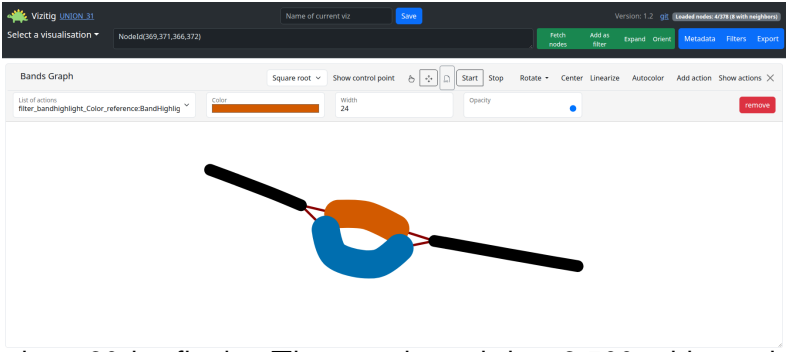
```
GGCTGATCAATGTGGCTATACCTTAGCAACATTTTTATGGGGTCGAATGGTTGATTCCTTT
|||||||||||||||||||||||||||||| ||||||||||||||||||||||||||||||
GGCTGATCAATGTGGCTATACCTTAGCAACTTTTTTATGGGGTCGAATGGTTGATTCCTTT
```



Compare both branch sequences with the aligner.

Union graph: an insertion

```
vizitig create "$INPUT_FOLDER" -k 31 -f 1 -n UNION
--edge-annotation
```



Both paths share 30 bp flanks. The sample path has 3,500 additional bp.